

Research Method of Data Science

Week 6: Qualitative Evidence and Analysis

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Week 6: Qualitative evidence and analysis

Tuesday: understanding how interpretation is built

Thursday: practicing and documenting an analysis

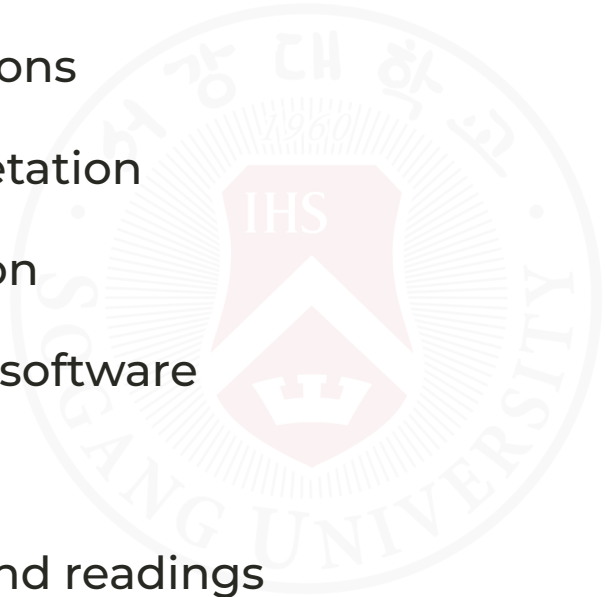
This week's output: an **individual qualitative research exercise**.



Agenda

Tuesday — Week 6, Meeting 1

1. Preparing qualitative evidence
 2. Coding and documenting decisions
 3. Categories, themes, and interpretation
 4. Trustworthiness and triangulation
 5. Mixed-methods integration and software
 6. Preparation for Thursday
- Appendix: lab corpus, templates, and readings



Week 5 connection

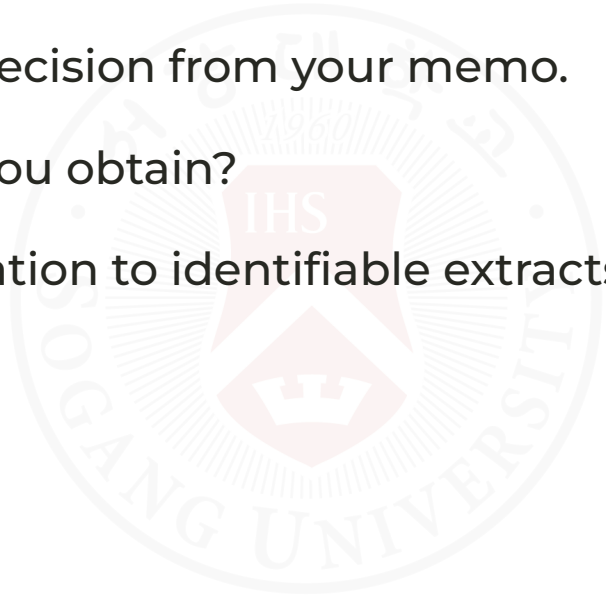
Last week: **Who or what will you study, and how?**

This week: **How will you turn that evidence into an answer?**

With a partner, recall one design decision from your memo.

How might it affect the evidence you obtain?

By Thursday: connect an interpretation to identifiable extracts, explain your decisions, and state the limits of your claim.



Our running example

Research question

How do university students decide when to use generative AI while studying?

Evidence: fictional interview extracts about study practices, plus a fictional course-policy document.

Possible contribution: understanding the conditions under which students describe AI as support or as a substitute for effort.

We are examining **accounts and meanings**, not measuring an effect on learning.

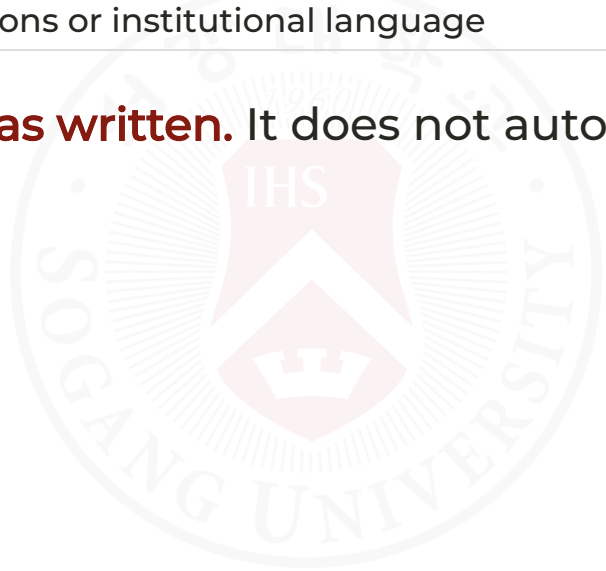
The background of the slide features a large, faint, circular watermark of the Seoul National University (SNU) logo. The logo contains the university's name in Korean and English, the year 1960, and the letters IHS.

1. Preparing qualitative evidence

What counts as qualitative evidence?

Material	What it may reveal	What needs checking
Interviews / focus groups	experiences and expressed meanings	prompts, audience, interaction
Observation notes	actions in a particular setting	observer role and omissions
Documents	official positions or institutional language	author, purpose, date, audience

A document is evidence of what was written. It does not automatically show what people actually do.



Transcription is an analytical decision

Transcription: representing recorded speech in written form.

- For an interview about study practices, readable verbatim text may be suitable.
- For interaction or discourse analysis, pauses, overlap, and delivery may matter.
- Preserve speaker labels, question context, and stable extract IDs.
- Mark uncertainty: **[inaudible 02:14]**; do not invent missing words.

Translation: retain the original, identify translated quotations, and document difficult terms.

Research Interview Transcript Example

Interview Title: Exploring the Impact of Technology on Education

Interview Date: [Insert Date]

Interviewee: [Interviewee's Full Name and Role/Position]

Interviewer: [Interviewer's Full Name]

Interview Location: [Physical/Online Platform]

Recording Method: [Audio/Video]

Transcript Style: Verbatim/Intelligent Verbatim/Edited

[Timestamp]

Interviewer: Can you describe how technology has influenced the way students learn in recent years?

Interviewee: Absolutely. Over the past decade, technology has transformed learning by making it more interactive and accessible. For instance, online platforms like [platform name] have enabled students to access resources anytime, anywhere.

[Timestamp]

Interviewer: What challenges do educators face when integrating technology into their classrooms?

Interviewee: One major challenge is ensuring equitable access to devices and the internet. Another is training teachers to use new tools effectively.

[Timestamp]

Interviewer: How has technology improved collaboration among students?

Interviewee: Tools like [tool name] and [tool name] allow students to work together on projects

Transcript challenge – 3 minutes

Fictional original speech:

| I don't trust it for facts. I do use it to suggest an outline.

Automatic transcript:

| I trust it for facts. I don't use it to suggest an outline.

1. Which interpretations would change?
2. What would you check before coding?
3. If you cannot verify the recording, how would you report this?

A traceable evidence file

Keep separate copies of **original**, **checked**, and **analysis** materials.

Field	Example
Source and extract ID	P01, E01
Context	first-year student; fictional interview
Preparation record	transcript checked; names replaced
Version	checked_v1; coded_v2
Decision note	removed location, retained relevant context

Replacing names may leave indirect identifiers. Keep any re-identification key separately and securely.

P01 = fictional speaker 1; E01 = extract 1; checked_v1 = first verified transcript; coded_v2 = first coding version

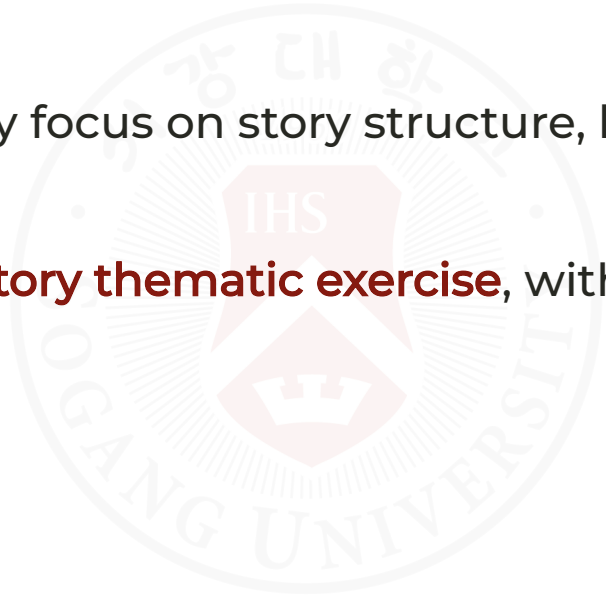
Choosing an analysis approach

Thematic analysis: develops patterns of meaning across evidence.

Qualitative content analysis: systematically categorises content; may also examine counts with suitable definitions.

Narrative or discourse analysis: may focus on story structure, language, or interaction rather than cross-source themes.

Our lab uses a **structured introductory thematic exercise**, with a revisable coding guide and peer discussion.



2. Coding and documenting decisions



Coding means labelling a relevant meaning

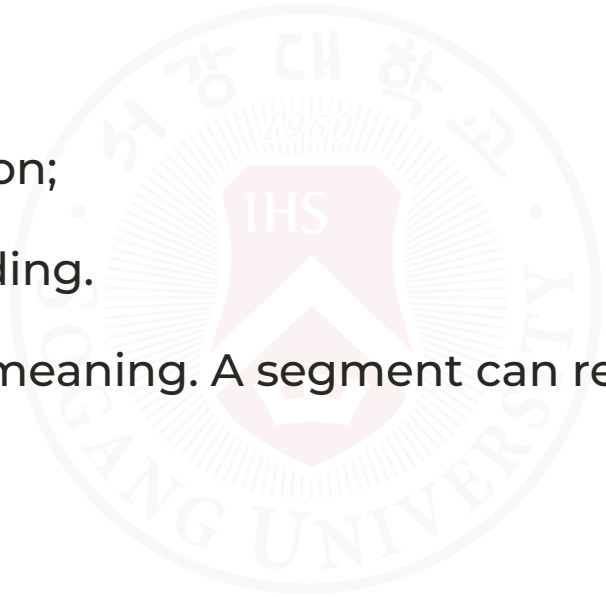
Code: a short label attached to an evidence segment.

| E01: “I ask AI to explain the lecture differently, then close it and try the problem myself.”

Possible codes:

- seeking an alternative explanation;
- testing independent understanding.

Code enough context to preserve meaning. A segment can receive more than one code.



Coding approaches can be semantic (surface meaning) or latent (underlying meaning).

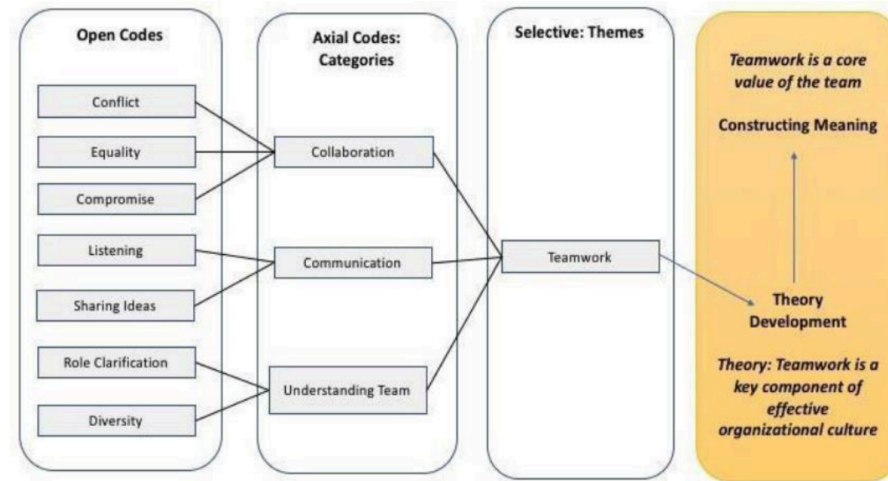


Figure 7. Creation of Theory and Meaning

This illustrates one organization of codes, categories, and theory. Coding procedures vary by analytical approach.

Inductive and deductive coding

Orientation	Starting point	Example
Inductive	ideas noticed while reading	“testing independent understanding”
Deductive	a question or conceptual framework	“self-monitoring” from a learning framework
Combined	initial concepts, revised through evidence	separate checking facts from checking understanding

Even inductive coding is shaped by the research question and the researcher's perspective.



Description and interpretation

E02: “When the deadline is close, I paste the question and use the first answer. I tell myself I'll understand it later.”

Semantic code: using the first answer under time pressure.

Possible interpretive code: postponing learning to complete the task.

Too far: “This student is lazy and cannot think critically.”

An interpretive reading needs contextual support and consideration of alternatives.

Try coding one extract – 5 minutes

E03: “I check its citations because once a paper didn't exist. But if the explanation sounds clear, I usually don't check that.”

Alone — 2 minutes

1. Mark one or two meaningful segments.
2. Give each a specific code.

Pairs — 2 minutes: explain your labels and one alternative.

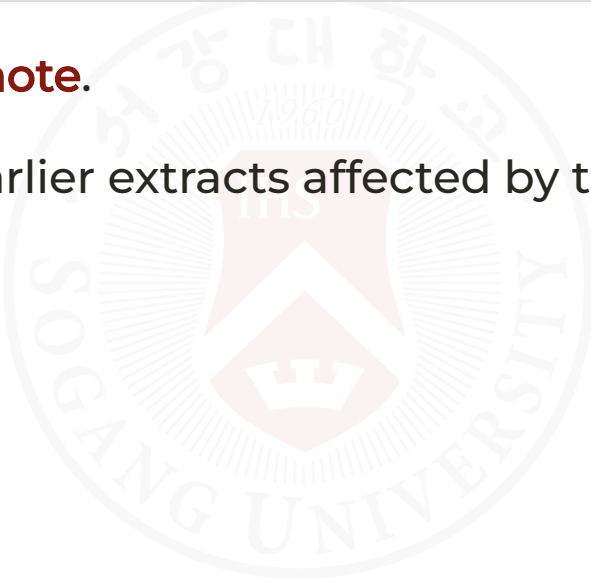
Class — 1 minute: what distinction would “AI trust” hide?

A useful coding guide

Code	Meaning / include	Exclude
Checking factual accuracy	compares an AI claim or source with another source	accepting fluent wording
Testing own understanding	attempts or explains a task independently	only comparing two AI answers

Add an **example ID** and a **revision note**.

If you change a definition, revisit earlier extracts affected by that change.



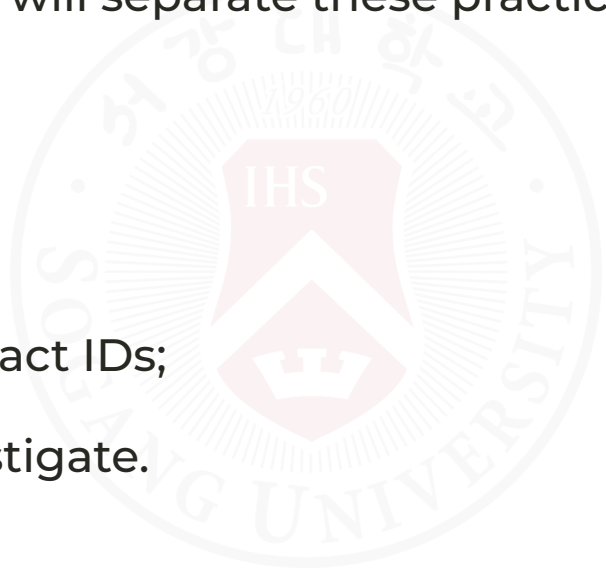
Memos record your reasoning

An **analytical memo** is a short note about an interpretation or decision.

“E01 checks understanding through an independent attempt, while E03 checks citations but trusts clear explanations. I will separate these practices rather than code both simply as ‘verification’.”

Record:

- what changed and why;
- supporting and challenging extract IDs;
- a question you still need to investigate.



3. Categories, themes, and interpretation



Codes, categories, and themes

Level	Job	Running example
Code	labels a segment	testing own understanding
Category	groups related codes	checking practices
Theme	develops a pattern of meaning answering the question	students apply different standards of checking to sources and explanations

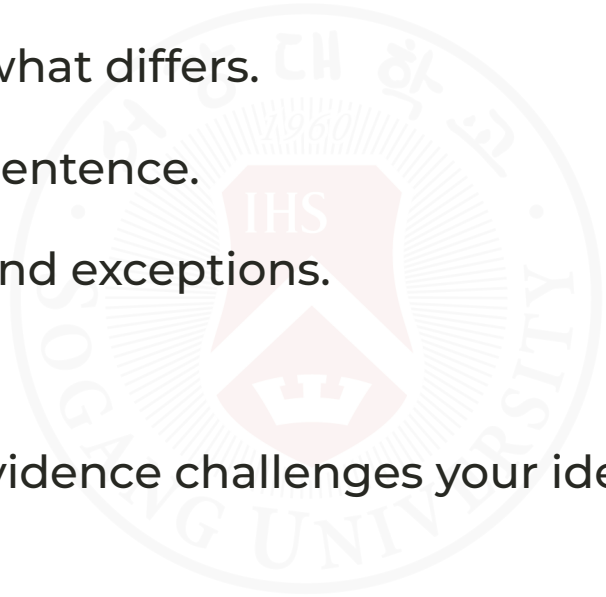
A theme explains a pattern; it is more than a folder called **“AI use.”**

Developing a theme

Read across extracts, not only within one speaker's account.

1. Bring related coded segments together.
2. Ask what connects them—and what differs.
3. Write a provisional pattern as a sentence.
4. Recheck it against full extracts and exceptions.
5. Refine its scope and name.

Return to earlier steps when the evidence challenges your idea.



Theme challenge – 5 minutes

Using E01–E03, compare these candidates:

A. “Students use AI.”

B. “Students manage the boundary between completing a task and understanding it.”

C. “AI destroys independent thinking.”

In pairs, choose the most promising candidate and:

- name supporting extracts;
- identify what it fails to capture;
- revise one word or phrase to limit the claim.

Writing a qualitative finding

Pattern → evidence → interpretation → qualification

These fictional accounts suggest that students distinguish task completion from independent understanding. E01 describes closing AI before attempting the problem; E02 postpones understanding to meet a deadline. This contrast suggests that time pressure may shape the boundary students describe. The extracts do not establish actual learning outcomes or how common these practices are.

A quotation supports an argument. It does not replace your explanation.

Claim boundaries

These extracts can support claims about **reported practices**.

They cannot establish:

- how many students in the university behave this way;
- whether AI caused a change in critical thinking;
- whether described checking actually occurred;
- whether one student's experience represents a country.

Counts of excerpts are not counts of independent people.

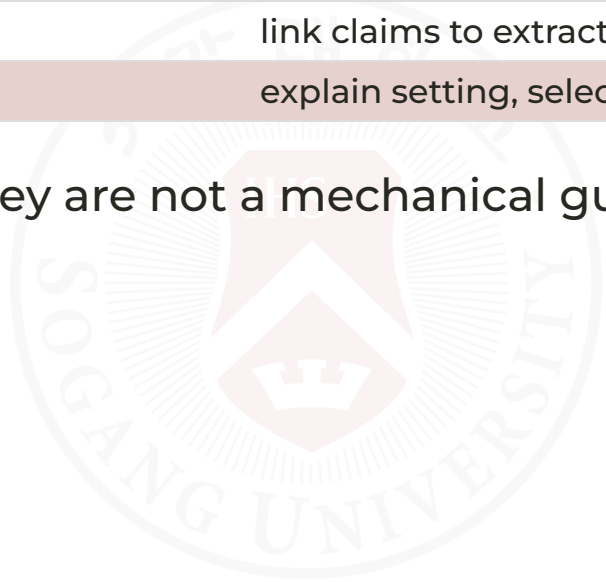
The seal of Chung-Ang University is a circular emblem. It features a central shield with a crown on top and the letters 'IHS' inside. The shield is set against a background of a circular border containing the university's name in Korean and English, along with the year '1960'.

4. Trustworthiness and triangulation

What makes an interpretation trustworthy?

Question	Practice
Is it credible?	check context, rival readings, and exceptions
Is the process traceable?	retain coding revisions and memos
Is it grounded in evidence?	link claims to extracts and decisions
Can readers judge relevance elsewhere?	explain setting, selection, and limits

These practices support quality. They are not a mechanical guarantee of truth.



An exception can improve your analysis – 4 minutes

Provisional claim: **“Deadlines make students accept AI answers without checking.”**

E04: “When I have very little time, I avoid AI because checking a plausible wrong answer takes longer than doing the task myself.”

Pairs: What happens to the claim?

- Discard E04 as inconvenient?
- Explain an important difference?
- Narrow or replace the theme?

Write one revised sentence.

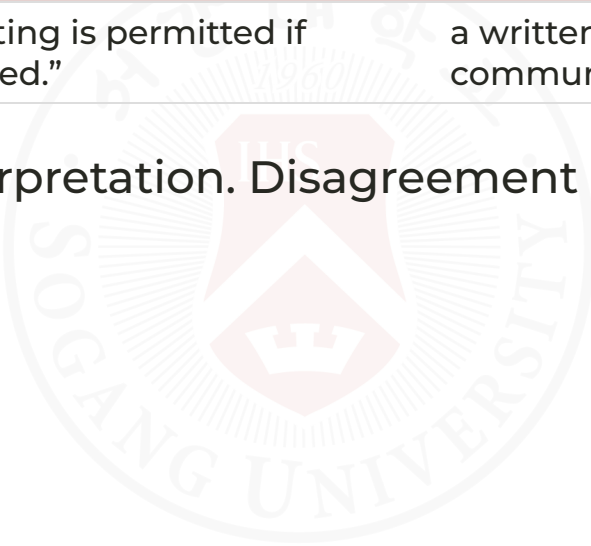


Triangulation: comparing perspectives

Triangulation examines evidence from different sources, methods, investigators, or theoretical perspectives.

Interview account	Course document	Interpretation
"I was unsure whether editing was allowed."	"AI editing is permitted if disclosed."	a written rule may not be understood or communicated clearly

Agreement can strengthen an interpretation. Disagreement may reveal something important.



Reflexivity during analysis – 3 minutes

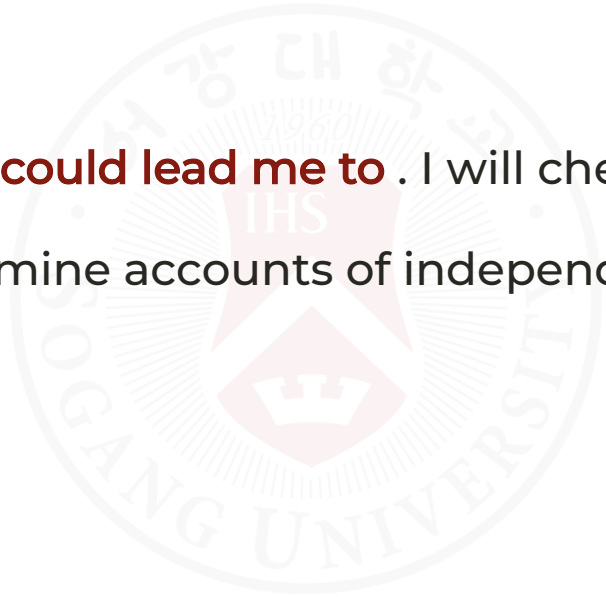
Imagine you strongly believe that **AI reduces student effort**.

What might you notice first? What might you overlook?

Privately write:

| “My initial expectation was . **It could lead me to** . I will check this by ____.”

A useful response: deliberately examine accounts of independent practice and alternative explanations.



5. Mixed-methods integration and software

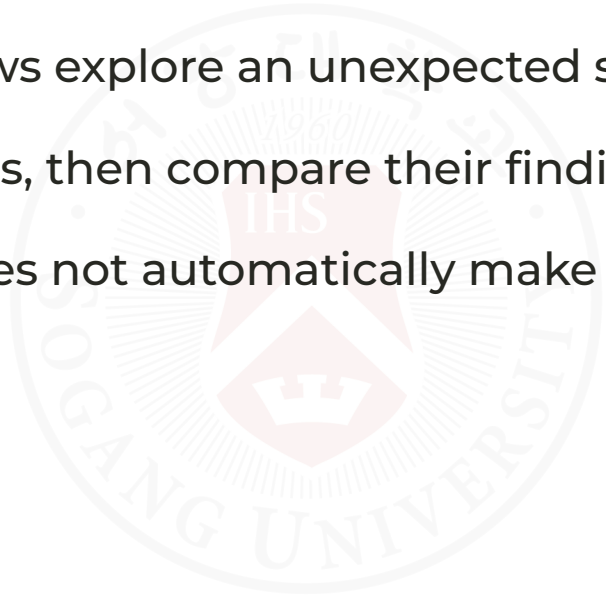


Mixed methods require integration

Qualitative and quantitative evidence must inform a **shared research question**.

- **Exploratory sequential:** interviews help develop a survey.
- **Explanatory sequential:** interviews explore an unexpected statistical pattern.
- **Convergent:** analyse both strands, then compare their findings.

Counting words or codes alone does not automatically make a study mixed methods.



A joint display connects findings

Hypothetical example—not observed results

Quantitative strand	Qualitative strand	Integrated interpretation
AI-use frequency has a weak association with test scores	students describe both explanation-seeking and answer-copying	frequency alone may combine different practices
association varies by preparation	some describe checking as easier with subject knowledge	preparation may help explain variation; a mechanism needs further testing

What new question follows? **Measure purposes and checking practices, not only frequency.**

Software options

Tool	Useful role	For this course
Word + Excel / Sheets	annotate text; record extracts, codes, memos	required workflow can use these
Taguette	highlight/tag documents and retrieve extracts	optional free/open-source route
QualCoder	code text/media and write memos	optional free/open-source desktop route
NVivo, MAXQDA, ATLAS.ti	manage coding, queries, retrieval and projects	specialist alternatives; check access

Software organises evidence. The researcher must justify the interpretation.

The same workflow in different tools

Word/Excel: read text → copy exact extract + ID → add code and memo → filter related codes → inspect full context.

Taguette: import a document → define tags → highlight a passage → apply tags → retrieve tagged passages → export work.

QualCoder or specialist software: follow the same analytical logic; menus and exports differ.

Always keep a readable export with **source IDs, quotations, codes, and decisions.**

AI assistance: suggestions need checking

AI can suggest transcript corrections, codes, or alternative readings.

For any suggestion you use:

1. Check it against the **exact extract and context**.
2. Record what you accepted, changed, or rejected.
3. Keep responsibility for your analysis and disclosure.

For Thursday, **complete your own coding first**. External AI uploads of real participant material require appropriate permission and safeguards.

Preparation for Thursday

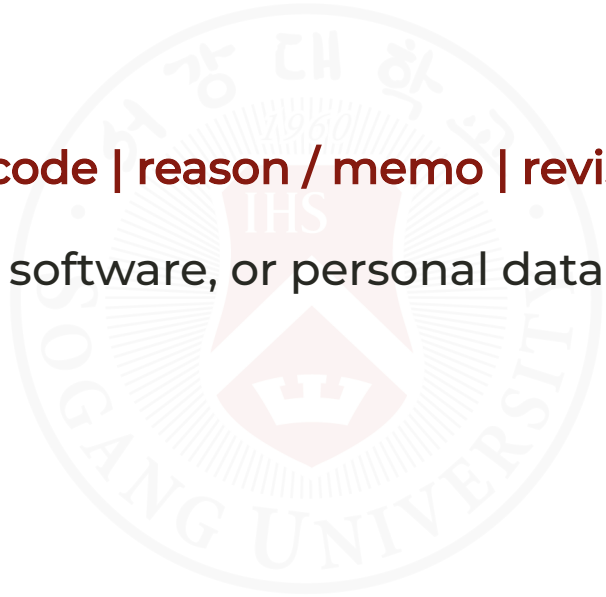
Bring a laptop with **Word and Excel**, or an equivalent text editor and spreadsheet.

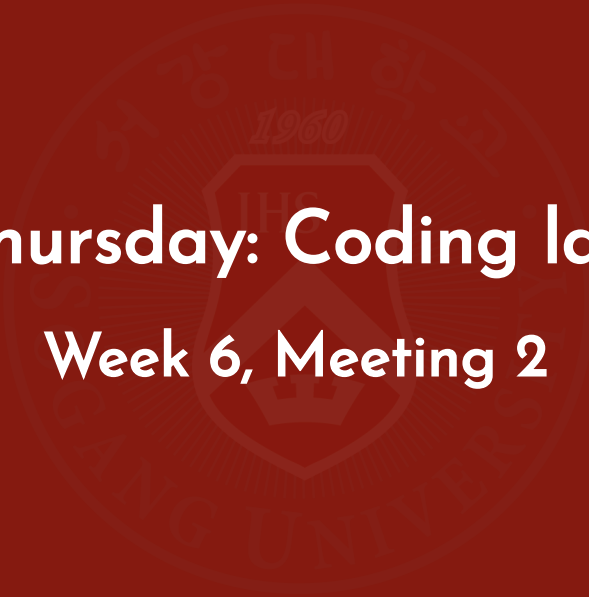
Read the **fictional lab corpus in the appendix** once.

Create a blank table:

Extract ID | exact segment | initial code | reason / memo | revised code

No new interviews, recording, paid software, or personal data are needed.





Thursday: Coding lab

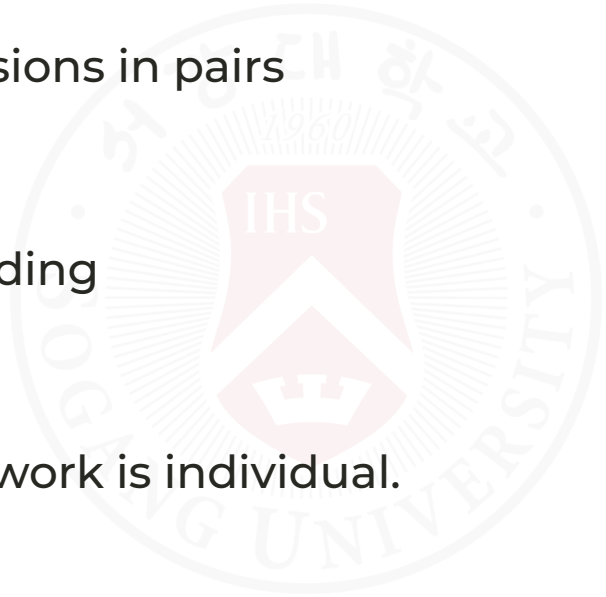
Week 6, Meeting 2

Agenda

Thursday — Week 6, Meeting 2

1. Read the evidence and establish context
2. Code individually; compare decisions in pairs
3. Develop and challenge themes
4. Write and revise a qualitative finding
5. Submit the individual exercise

10 students → five pairs. Your final work is individual.



Today's individual output

One analysis pack containing:

- coded evidence table for E01–E12;
- a working guide with 3–6 codes;
- two provisional themes with supporting/challenging IDs;
- a 150–200-word finding;
- a short decision/reflexivity note and AI-use statement.

The fictional policy D01 helps you examine a source discrepancy.

Starting your evidence table – 3 minutes

1. Open a blank spreadsheet or Word table.
2. Add the five column headings from Tuesday.
3. Preserve E01–E12 and P01–P06 identifiers.
4. Save as `week6_yourname_analysis`.

What you should see: one row per segment, with space for a code and memo.

If one extract has two distinct meanings, use two rows with the same ID.

1. Familiarisation – 8 minutes

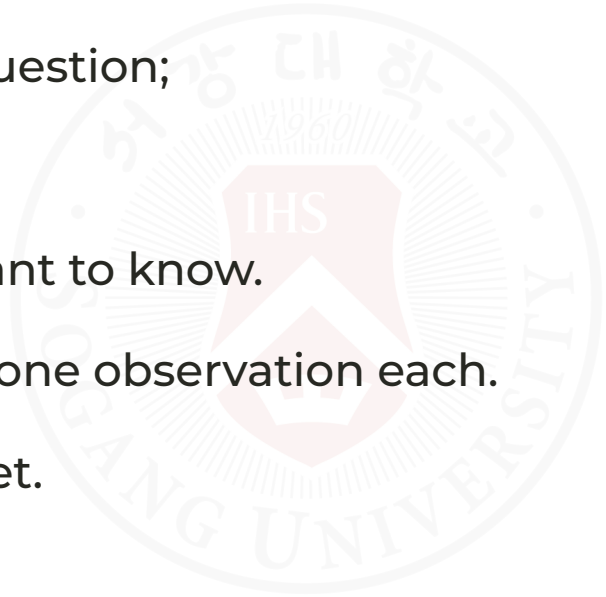
Read **all six speakers' accounts and D01** before coding.

Alone — 5 minutes: jot down:

- what seems important for the question;
- a tension or contradiction;
- one missing detail you would want to know.

Class — 3 minutes: students share one observation each.

Do not choose your final themes yet.



2. Initial coding – 15 minutes

Alone — 10 minutes

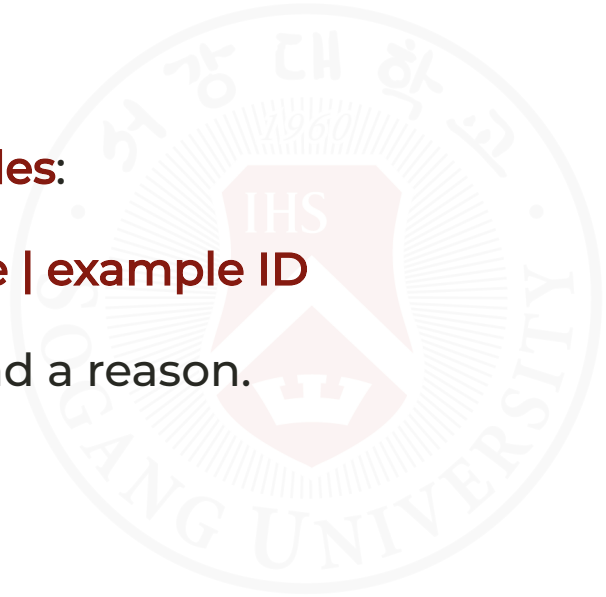
Code E01–E12 with specific labels. Preserve exact words and relevant context.

Then — 5 minutes

Draft a guide with **3–6 working codes**:

name | meaning | include / exclude | example ID

Mark uncertain decisions with **?** and a reason.



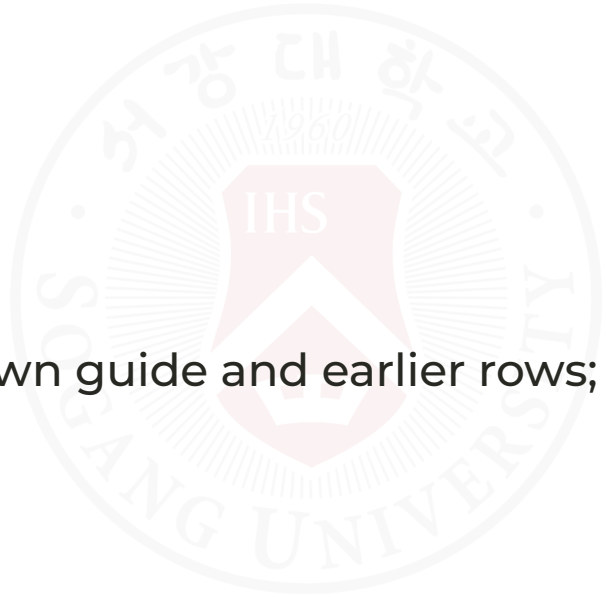
3. Pair comparison – 12 minutes

Compare — 5 minutes: select two extracts where your interpretations differ.

Discuss — 4 minutes: is the difference about:

- the segment selected;
- the code's definition;
- a competing interpretation?

Revise — 3 minutes: update your own guide and earlier rows; record one changed or retained decision and why.



4. Developing themes – 14 minutes

Alone — 8 minutes: build two provisional themes.

For each, record:

- a sentence stating its pattern of meaning;
- at least two supporting extracts;
- an exception, tension, or limit;
- how it answers the research question.

Pairs — 4 minutes: challenge one theme.

Class — 2 minutes: one pair explains a revision.

A theme needs a boundary

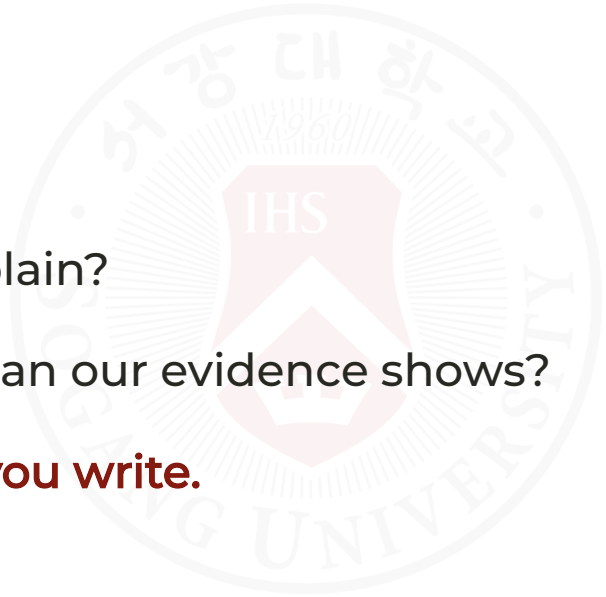
Too broad: “AI has advantages and disadvantages.”

More analytical: “Students describe verification as a cost that changes when AI is worth using.”

Ask:

- Which extracts support this?
- Which practice does it fail to explain?
- Does the wording claim more than our evidence shows?

Theme names may change while you write.



5. Interpretation – 10 minutes

Write **150–200 words** answering the running question.

Include:

1. One clear thematic claim.
2. Two exact short quotations with IDs.
3. Your explanation of their relationship.
4. One contrasting extract or rival interpretation.
5. One comparison with D01 and a claim limitation.

State explicitly: **this is analysis of fictional teaching data.**

6. Peer feedback and revision – 8 minutes

Exchange findings with your partner.

Read — 2 minutes: underline the claim and the evidence.

Discuss — 3 minutes:

- Can the claim be traced to the extracts?
- Does the text interpret or only list quotations?
- Is an exception or limitation handled honestly?

Revise — 3 minutes: change your own paragraph and record why.

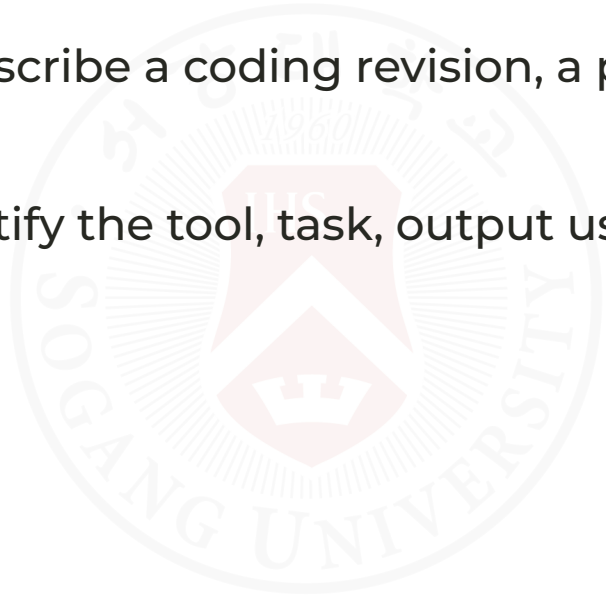
7. Submission – 3 minutes

Submit through **Cyber Campus** at the deadline announced there.

Check: coded table; guide; themes; finding; decision note; AI statement.

Decision note — 3–4 sentences: describe a coding revision, a possible personal assumption, and a limitation.

AI statement: “No AI used,” or identify the tool, task, output used, and verification.



Connection to your complete research paper

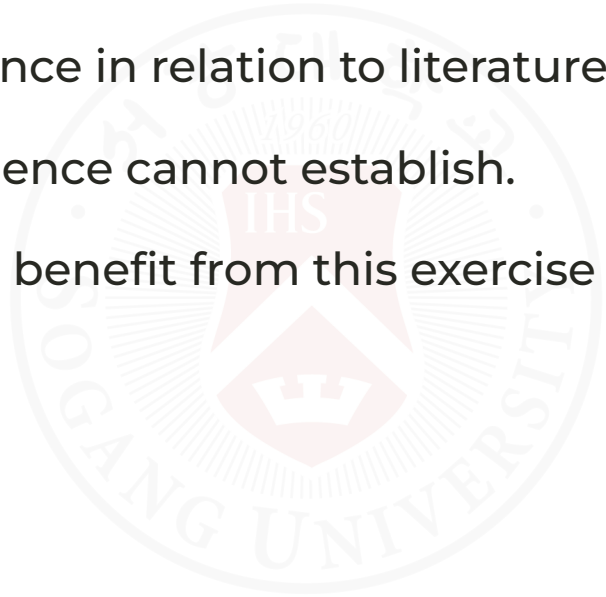
Methods: explain evidence selection, preparation, analysis, ethics, and researcher role.

Results: present patterns with traceable evidence.

Discussion: interpret their significance in relation to literature.

Limitations: explain what your evidence cannot establish.

A quantitative final project can still benefit from this exercise in interpretation and research judgment.



The background of the slide features a large, faint, circular seal of Yonsei University. The seal contains the university's name in Korean and English, the year 1960, and the letters IHS.

Appendix

Lab corpus, templates, and reference materials

Lab corpus: context

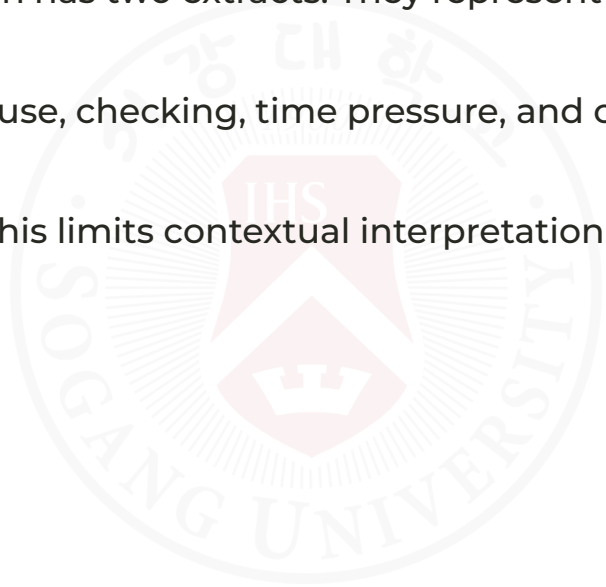
All participants, quotations, and policy text are fictional.

Question: **How do university students decide when to use generative AI while studying?**

P01–P06 are six invented student speakers. Each has two extracts. They represent contrasting accounts for practice, not a sample of actual Sogang students.

The imagined interview asked about recent AI use, checking, time pressure, and course expectations. D01 is an imagined written course policy in the same setting.

We have selected extracts, not full interviews; this limits contextual interpretation.



Lab corpus: P01 and P02

P01 — E01

| I ask AI to explain the lecture differently, then close it and try the problem myself.

P01 — E05

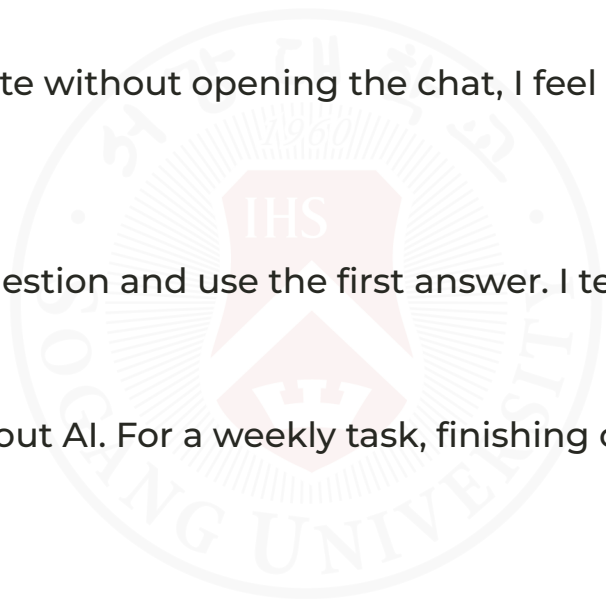
| If I can explain the answer to my roommate without opening the chat, I feel I've understood it. Sometimes I discover I can't.

P02 — E02

| When the deadline is close, I paste the question and use the first answer. I tell myself I'll understand it later.

P02 — E06

| For exam revision I do the questions without AI. For a weekly task, finishing can matter more to me than practicing.



Lab corpus: P03 and P04

P03 — E03

| I check its citations because once a paper didn't exist. But if the explanation sounds clear, I usually don't check that.

P03 — E07

| In a subject I know, I notice mistakes. In a new subject, I don't really know what I should check.

P04 — E04

| When I have very little time, I avoid AI because checking a plausible wrong answer takes longer than doing the task myself.

P04 — E08

| I use it after I have a draft, to suggest a different structure. I reject suggestions that change what I meant.

Lab corpus: P05 and P06

P05 — E09

One teacher said AI was fine for editing. Another just said don't rely on it. I wasn't sure where editing ended and doing the work began.

P05 — E10

I started writing down what I used it for. That helped me notice when I had accepted more than grammar changes.

P06 — E11

I ask two AI tools the same question. When they agree, I usually assume it's right; I don't always open another source.

P06 — E12

Once both gave the same wrong answer. Now I compare factual claims with the textbook, but I still use them for brainstorming.

Lab corpus: fictional document D01

Imagined course policy

Students may use AI to suggest explanations, brainstorm ideas, or edit language. They must verify factual claims and disclose use. Submitting an AI answer as one's own work without substantive independent engagement is not permitted.

Source type: written policy, not participant testimony.

Compare D01 with E09–E10. What differs between a formal rule and a student's account of its meaning?

Evidence table template

ID / speaker	Exact segment	Initial code	Reason / memo	Revised code
E01 / P01	[copy exact words]	[your label]	[why this label?]	[retain or revise]
...	...	...	...	...

Coding guide

Code	Meaning	Include / exclude	Example ID	Revision
[name]	[definition]	[boundary]	[ID]	[change + reason]

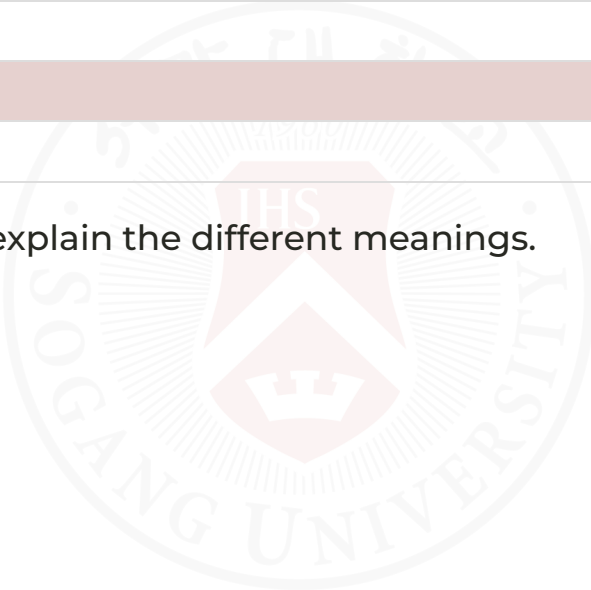
Keep participant quotation and your interpretation in separate columns.

Theme worksheet

Field	Theme 1	Theme 2
Provisional name		
Pattern in one sentence		
Supporting IDs / speakers		
Tension, exception, or boundary		
Connection to question		

One extract can inform two themes when you explain the different meanings.

Do not count it as two independent people.



Optional AI audit after independent analysis

Use **fictional text only** and record the tool/date.

Suggested prompt:

Suggest two possible codes for E07. Quote the exact segment supporting each. Offer an alternative reading and say what cannot be inferred. Do not invent context.

Compare its suggestions with your own. Log one acceptance, revision, or rejection and the evidence behind it.

No AI is required to complete the exercise.

Another example: central-bank documents

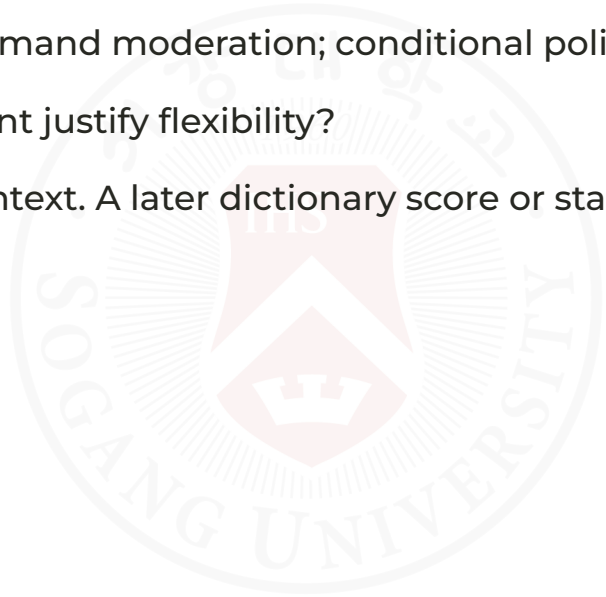
Invented statement:

Inflation remains above target, but tighter financing conditions are slowing demand. Future decisions will depend on incoming data.

Possible codes: persistent inflation concern; demand moderation; conditional policy stance.

Interpretation question: how does the statement justify flexibility?

Qualitative reading examines meaning and context. A later dictionary score or statistical analysis answers a different question and needs validation.



Another example: public policy documents

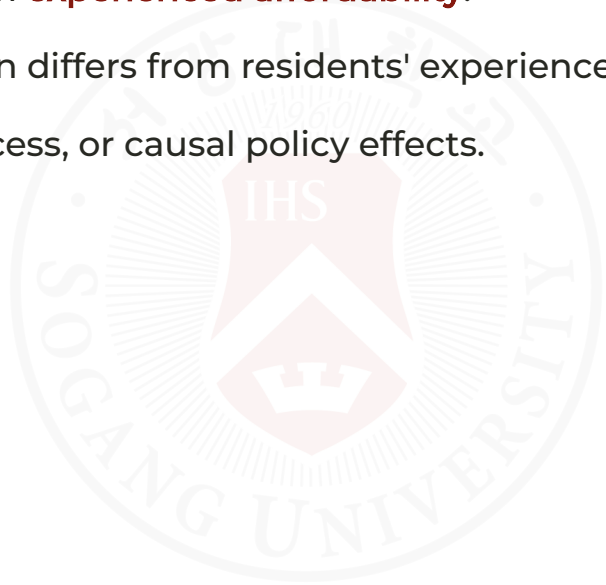
Invented municipal report: “New housing units will improve affordability.”

Invented resident account: “More units opened, but I still cannot afford the rent.”

Codes might distinguish **supply expansion** from **experienced affordability**.

A theme could explore how an official definition differs from residents' experiences.

Further evidence is needed to judge prices, access, or causal policy effects.



Reference: thematic analysis

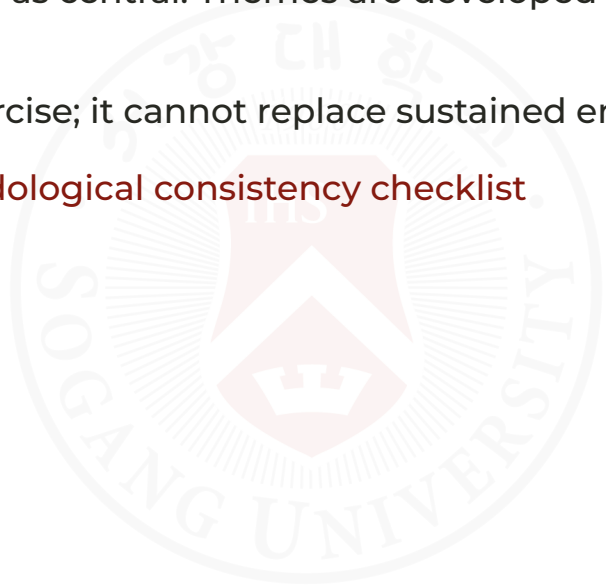
Braun and Clarke's reflexive thematic analysis describes a recursive process:

familiarisation; coding; initial themes; developing/reviewing themes; refining/naming themes; writing.

Their approach treats researcher interpretation as central. Themes are developed through analysis rather than found automatically.

Our short lab is an introductory structured exercise; it cannot replace sustained engagement with a real dataset.

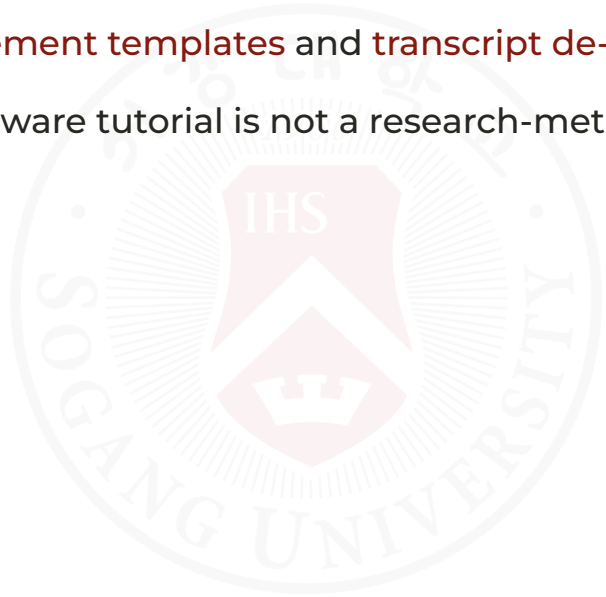
Understanding TA · Doing reflexive TA · Methodological consistency checklist



Reading and practical resources

- Braun & Clarke, **Thematic Analysis**: official explanations and teaching resources at thematicanalysis.net.
- Feters, Curry & Creswell (2013), **Achieving Integration in Mixed Methods Designs—Principles and Practices**, *Health Services Research*, 48(6 Pt 2), 2134–2156. [Article](#).
- UK Data Service: [transcription/data-management templates](#) and [transcript de-identification](#).

Choose reading relevant to your method; a software tutorial is not a research-method justification.



Software guides

Taguette getting started: importing, tagging, retrieval, exports.

QualCoder overview: text/media coding and memos.

NVivo · MAXQDA · ATLAS.ti: official product information and help.

Check current access, operating-system support, export options, and data-handling arrangements before choosing a tool.

Required lab route remains Word/Excel or equivalent.

